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# PROJECT :
## Smart Greenhouse Automated with Arduino

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# Foreword

This project is part of our C++ training and aims to put into practice the concepts studied in object-oriented programming in C++ in order to consolidate our knowledge and develop concrete skills.

We would like to express our deep gratitude to our professor, Mrs. Rim Megdiche, for her valuable supervision, her wise advice and her availability throughout the completion of this project. Her support and recommendations greatly contributed to the quality of our work and to our learning.

We hope that this project reflects the seriousness and the commitment that we devoted to its realization.

# Introduction

Mastering environmental conditions is today an essential issue in the fields of modern agriculture, sustainable food production and the protection of natural resources. Indeed, climate variability, irregular rainfall, rising temperatures and the decrease in arable land are increasingly pushing towards the adoption of automated systems capable of optimizing plant growth.

The central problem lies in the difficulty of manually maintaining parameters such as temperature, humidity or even air quality, which directly influence the yield and health of crops. Traditional methods often prove imprecise, time-consuming and dependent on human intervention, which limits efficiency and increases the risk of errors.

Faced with these constraints, the integration of embedded systems and sensor technologies is a promising solution for automating the monitoring and regulation of growing conditions.

It is in this context that our smart greenhouse project fits, based on the use of the Arduino board, sensors and various actuators, programmed in C++. The objective is to propose a functional model capable of measuring environmental parameters in real time and acting automatically when the defined thresholds are no longer respected, for example by activating a ventilation system when the temperature exceeds a critical value, or by adjusting humidity through a spraying module.

Thus, this project aims not only to provide a technological response to a real problem but also to develop technical skills such as microcontroller programming, the processing of data from sensors and the design of a complete automated system.

Our greenhouse model will constitute a proof of concept showing how embedded electronics can contribute to a more efficient, smarter and more sustainable agriculture. This work also paves the way for future improvements such as the integration of IoT communication, remote control or even predictive analysis of plant needs.

# Context – Presentation of the structure

## 1.1 General presentation of the project

The project is developed in a university setting by a team of students working on object-oriented C++ programming.

The objective of the structure is to design educational solutions based on electronics and computer science, in order to experiment with real systems.

The connected greenhouse project fits into this learning logic, by offering a platform that makes it possible to automate the monitoring of environmental conditions (temperature, humidity, watering) using Arduino and various sensors.

This work contributes to the development of intelligent solutions aimed at energy optimization and automated management.

## 1.2 System Objectives

The system aims to:

- Monitor the state of a greenhouse in real time
- Automate certain actions (watering, temperature control)
- Improve plant growth conditions
- Provide a reusable technical base for other projects (IoT, smart agriculture).

## 1.3 System target

This project is aimed at:

- Students and teachers in electronics, computer science and automation
- Gardening enthusiasts wishing to automate their greenhouse
- Small agricultural structures looking for a simple and inexpensive solution.

The main needs of the target are ease of use, reliability and the ability to consult data easily.

## 1.4 Quantative objectives

- Continuously measure temperature, soil humidity and air humidity.
- Automate watering as soon as a critical threshold is reached.
- Ensure stable operation 24/7.
- Allow the display or recording of measurement data.

### 1.5 Project scope

The system includes:

- An Arduino board;
- Sensors (temperature, humidity, soil);
- Actuators (pump, fan, LED, etc.);
- C++ code ensuring processing and automation.

The project does not provide for a multilingual version or a complex web interface: the emphasis is on the hardware and embedded software operation.

### 1.6 Description of the existing situation

The available resources are: the physical model of the greenhouse, the sensors, the electronic components and the initial C++ code. The team has the necessary technical skills and the equipment for assembly and testing.

# General description of the system

The project consists of designing and building an automated watering and climate control system intended for a mini-greenhouse.

- The mechanism is based on two water pumps: a 5V pump, submerged in a first reservoir, and a 12V pump, connected to a second reservoir. These two pumps are controlled using two relays ensuring safe and sequential operation.

- The first reservoir contains the water needed for the watering cycle. The role of the 5V pump is to transfer water from the first reservoir to the second. Then, the 12V pump draws this water and directs it to a funnel, whose flow moistens a capacitive humidity sensor, thus simulating the watering of soil. This sensor makes it possible to verify that the humidity is sufficient to guarantee correct watering. After a few seconds, a 220V heating system, controlled by a relay, is activated in order to dry the sensor and allow the experiment to be repeated. The heating also simulates hot weather in summer or the temperature surge in winter. This complete cycle is programmed to repeat three times.

- A water level sensor placed in the first reservoir ensures that the quantity of water is sufficient before the system is triggered. Without available water, the cycle does not start, guaranteeing the safety of the equipment.

- Outside the greenhouse, a DHT11 sensor measures the ambient temperature and humidity. This data is displayed in real time on a Liquid Crystal Display (LCD). When the outside temperature falls below 20°C, the heating system activates automatically to warm the environment. If the temperature exceeds 30°C, a 12V fan, also controlled by a relay, is triggered to cool the space. The role of the system is therefore to maintain the temperature within a stable range between 20°C and 30°C.

- A 16×2 LCD screen continuously displays the message:
« Project carried out by Ahmed Akrout and Khalil Basdouri – Supervised by Mrs. Rim Megdiche », which allows clear identification of the work.

- The device also includes three LEDs: a green LED indicating normal operation and remaining constantly on, a red LED blinking rapidly when the temperature goes outside the 20–30°C range, and a yellow LED that lights up while the 12V pump is running.

Finally, the entire system is programmed in C++ on Arduino, ensuring the coordinated management of the sensors, actuators and automatic sequences.

# Expected Features (Functional Requirements)

1) **Soil humidity measurement**
   - The soil humidity sensor provides an analog value.
   - If the value indicates that the soil is too dry → start the water pump.
   - When the humidity becomes sufficient → stop the pump.
2) **Air temperature and humidity measurement**
   - The DHT11 sensor is used.
   - Temperature and humidity are read every few seconds.
   - If the temperature exceeds a defined threshold (e.g.: 30 °C) → activate the fan.
   - When the temperature drops back down → stop the fan.
3) **Data display**
   - An LCD screen displays the air temperature and humidity in real time.
   - Soil humidity is displayed via the Arduino Serial Monitor.
4) **System safety**
   - The pump must never remain on longer than necessary.
   - An external push button prevents the 12V pump from operating if the box is open.
   - The thresholds must avoid any overheating or excessive watering.
   - The components must be powered within their limits (5V/12V).

**Non-functional requirements:**
These requirements do not directly concern the features, but the quality of the project.

- Reliability: the system must be able to operate for several hours without restarting.
- Code readability: the program must be well commented and organized.
- Ease of maintenance: the cables must be tidied to facilitate modifications.
- Energy saving: reasonable use of components.
- Electrical safety: no exposed wires, proper insulation, stable power supply.

# Technical constraints

**Programming language :** C++ (Arduino IDE).

**Class structures :**

→ **Sensor classes :**

TemperatureHumiditySensor : DHT11 management

WaterLevelSensor : Monitors the water in the reservoir

SoilMoistureSensor : Capacitive soil humidity sensor

→ **Actuator classes :**

WaterPump : Controls the 5V and 12V pumps

Heater : Controls drying and heating

CoolingFan : Temperature regulation

LED : With blinking capabilities

→ **Display classes :**

SevenSegmentDisplay : Temperature and humidity display

LCDDisplay : Display of the scrolling message

→ **Main system class :**

PlantWateringSystem : Orchestrates the entire system

**Mandatory hardware :**

- Arduino board
- DHT11 sensor (temperature and humidity)
- Capacitive soil humidity sensor
- Water level sensor
- 5V and 12V water pump
- Heating system
- 12V fan
- 5 electromagnetic relays to control the pumps, the heating and the fan
- 16×2 LCD screen

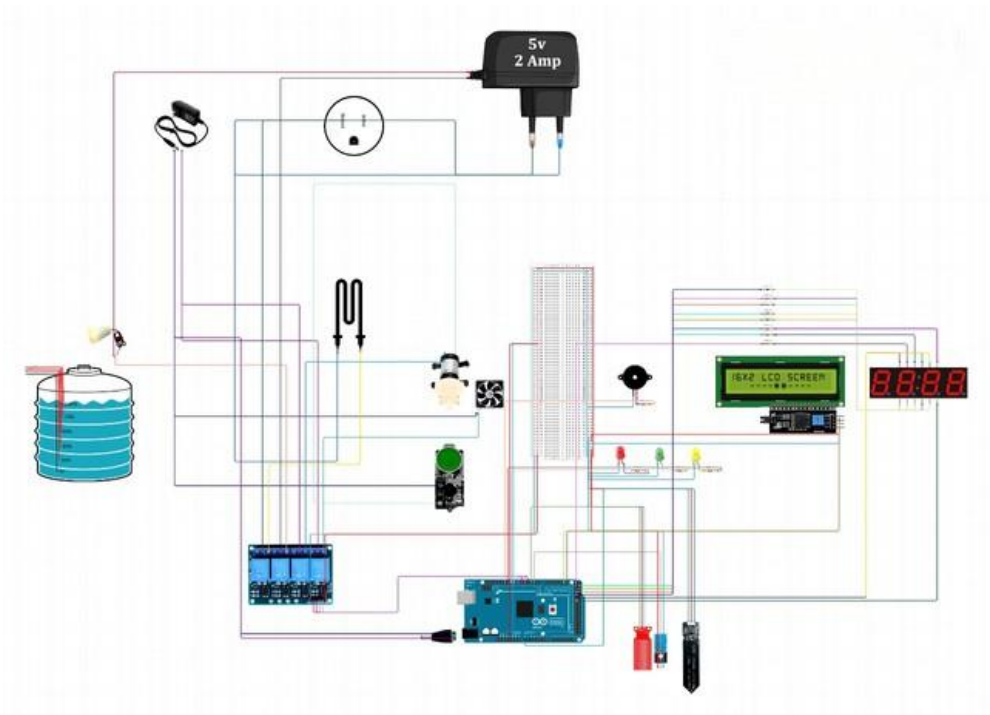
**Power Supply :**

- Arduino powered by USB or adapter,
- Actuators powered separately.

**Libraries used:**

- DHT.h (for the DHT)
- LiquidCrystal.h
- Wire.h (for I2C usage)

# Operating Principle Diagram




**Study of the wiring and assembly:**
The assembly was designed and simulated using the web application CirKit Designer in order to validate the operation before the actual realization. The heart of the system is an Arduino Mega 2560 board, which manages the sensors, the actuators and the display.

The sensors (water level sensor, capacitive soil humidity sensor and DHT11 sensor) are powered at 5 V and connected to the analog and digital inputs of the Arduino. The actuators (5 V and 12 V pumps, 220 V heating system and 12 V fan) are controlled via relay modules, allowing isolation between the low-voltage control part and the power part.

The 16×2 LCD screen are connected to the digital outputs of the Arduino, with limiting resistors to protect the components. The signaling LEDs (green, red and yellow) are also connected to the digital outputs to indicate the state of the system.

The power supply is divided between a 5 V source for the Arduino and the sensors, and a 12 V source for the pumps and the fan, with a common ground ensuring signal consistency. This wiring guarantees safe, modular operation in accordance with the constraints of the smart greenhouse project.

# Conclusion

This project made it possible to design and simulate an automated management system for a smart greenhouse, based on an Arduino board and a coherent set of sensors and actuators. The developed system ensures automatic watering of the plants, control of the water level, as well as regulation of the ambient temperature and humidity. The entire wiring and operation was validated by a simulation under CirKit Designer, guaranteeing hardware consistency before any physical realization.

On the technical level, the project implements fundamental notions of embedded systems such as data acquisition via sensors (DHT11, soil humidity sensor, level sensor), logical processing by microcontroller, and the control of power devices using relays. The display of information on an LCD screen improves readability and interaction with the user, while the light indicators reinforce the supervision of the system state.

From a pedagogical and scientific point of view, this work constitutes a concrete application of the concepts of automation, electronics and C++ programming. It opens up prospects for improvement, in particular through the integration of more precise sensors, wireless communication systems (IoT), or even intelligent algorithms allowing energy optimization and predictive management of environmental conditions. Thus, this project can be considered a solid basis for future developments in the field of smart agriculture and cyber-physical systems.

# Appendix





